Assignment 16 Introduction to Javascript

Q1. What are conditional statements? Explain conditional statements with syntax and examples.

Ans) Conditional statements are used in JavaScript to make decisions in the code. They allow the program to execute a block of code only if a specific condition evaluates to true. If the condition is false, another block of code can be executed.

In simple words, conditional statements help control the flow of execution based on conditions.

Syntax:

```
If(condition){}
```

Example:

```
let age = 18;
if (age >= 18) {
   console.log("You are eligible to vote.");
}
```

2. if...else statement

The if...else statement executes one block if the condition is true, and another block if it is false.

Syntax:

```
if (condition) {
   // code if condition is true
} else {
   // code if condition is false
}
```

Example:

```
let num = 5;
if (num % 2 === 0) {
   console.log("Even number");
} else {
   console.log("Odd number");
}
```

3. if...else if...else statement

Used when multiple conditions need to be checked.

```
if (condition1) {
   // code if condition1 is true
} else if (condition2) {
   // code if condition2 is true
} else {
   // code if none of the conditions are true
}
```

Example:

```
let marks = 75;
if (marks >= 90) {
   console.log("Grade: A");
} else if (marks >= 75) {
   console.log("Grade: B");
} else {
   console.log("Grade: C");
}
```

4. switch statement

The switch statement is used to execute one block of code out of many options, based on the value of an expression.

```
switch(expression) {
   case value1:
      // code block
      break;
   case value2:
      // code block
      break;
   default:
      // code block if no case matches
}
```

Example:

```
let day = 3;
switch(day) {
```

```
  case 1:
    console.log("Monday");
    break;
  case 2:
    console.log("Tuesday");
    break;
  case 3:
    console.log("Wednesday");
    break;
  default:
    console.log("Other day");
}
```

Question 2)

Write a program that grades students based on their marks^ 8 If greater than 90 then A GradE 8 If between 70 and 90 then a B gradE 8 If between 50 and 70 then a C gradE 8 Below 50 then an F grade

Answer) Answer.

Question 3) What are loops, and what do we need them? Explain different types of loops with their syntax and examples.

Answer)

In programming, loops are control structures that allow us to execute a block of code repeatedly until a certain condition is met.

Instead of writing the same code multiple times, we use loops to save time, reduce redundancy, and make programs more efficient.

1. for loop

Used when we know in advance how many times we want to run the block of code.

Syntax:

```
for (initialization; condition; increment/decrement) {
  // code to be executed}
```

Example:

```
for (let i = 1; i <= 5; i++) {
  console.log("Number: " + i);
}
```

2. while loop

Used when we **don’t know** in advance how many times the loop will run. The code executes **as long as the condition is true**.

Syntax:

```
while (condition) {
  // code to be executed
}
let i = 1;
while (i <= 5) {
  console.log("Number: " + i);
  i++;
}
```

3. do...while loop

Similar to while, but the code executes **at least once** (because the condition is checked after execution).

```
do {
  // code to be executed
} while (condition);
```

Example:

```
let i = 1;
do {
  console.log("Number: " + i);
  i++;
} while (i <= 5);
```

4. for...in loop

Used to iterate over the **properties of an object**.

Syntax:

```
for (let key in object) {
   // code to be executed
}
```

Example:

```
let person = {name: "John", age: 25, city: "Delhi"};

for (let key in person) {
   console.log(key + ": " + person[key]);
}
```

5. for...of loop

Used to iterate over **iterable objects** like arrays, strings, etc.

Syntax:

```
for (let element of iterable) {
   // code to be executed
}
let fruits = ["Apple", "Banana", "Mango"];

for (let fruit of fruits) {
   console.log(fruit);
}
```

Q4. Generate numbers between any 2 given numbers. Ex⍰ 8 const num1 = 10⍰ 8 const num2 = 25; Output: 11, 12, 13, …., 25

Answer 4) Answer

Question 5) Use the while loop to print numbers from 1 to 25 in ascending and descending order

Answer 5) Ascending Descending